

RESTAURANT SPENDING ANALYSIS

Plotly Express tips dataset | Python / R | Customer Analytics

Project objective

Examine restaurant spending patterns and relationships between bill size and visit characteristics, with a focus on day, smoker status, meal time and party size.

Data

The dataset contains 244 transactions and seven fields: total_bill, tip, sex, smoker, day, time and size. Bill category is created using tertiles of total_bill.

Analytical workflow

1. Descriptive statistics compare mean, SD and count of total_bill across groups.
2. Welch independent-samples t-test compares total_bill for smokers and non-smokers.
3. Chi-square test examines day versus bill_category.
4. OLS regression models total_bill as a function of party size.
5. Shapiro-Wilk and Breusch-Pagan diagnostics are used; HC3 robust SE provide a sensitivity check for heteroscedasticity.

Business interpretation

Party size is relevant to table allocation, expected spend and capacity planning. Group differences should be interpreted as associations, not causal effects.

Portfolio value

Demonstrates Python-based data cleaning, descriptive analysis, hypothesis testing, categorical association testing, linear regression and model diagnostics.

Reproducibility

The repository contains analysis.py and tips.csv. The supplied scripts reproduce the core descriptive and inferential workflow from the public CSV.